

Project Participants			
Name	Cobot AMR and 7 th axis control		
Stakeholders	Project Sponsor	Tecnológico de Monterrey	
	Customers	Bridgex	
	Contractors	None	
	Project Team and Project Manager		
Project Manager	Ixchel Orozco Viveros	Email Address or Website	A01644531@tec.mx
Assistant Project Manager	Emilio Medina Velázquez	Email Address or Website	A01644639@tec.mx
Project Team	Sebastián Rivera López Iván Delgado Ramos Gerardo Galaviz Becerra		
Project Description			
Goal Statement	Design, integrate and deliver a system of flexible manufacture composed of an Autonomous Mobile Robot (AMR), with an integrated robotic arm for handling, a cobot for measuring pieces with a profilometer, and another cobot mounted on a linear 3-meter-long 7th axis, where its position is controlled.		
Description and Background	The rapid growth of the manufacturing industry has increased the demand for solutions that increase the productivity of a process; and current technology allows flexible automated systems to adapt to the necessities and rapid change of the environment of an industry. We propose a flexible system that allows the automatic handling and measuring of pieces, with an integrated AMR, a robotic arm and two cobots.		
Objectives	Integrate a flexible manufacturing system that includes the following components: • AMR: For transport of pieces • Robotic arm: Handling of pieces • Cobots: Handling of profilometer and pieces • Pieces location and identification system • Automated pieces inspection sequence		
Scope	• Prototype of an AMR compliant with industry standards • 4 degrees-of-freedom robotic arm with an end effector. • Sequence of measurements with profilometer. • Control of position of the cobot mounted on the 7th axis with a PLC. • Control software for locating the piece, measuring it with the profilometer, and withdrawal of the piece.		
Deliverables	• AMR with intelligent navigation system, mechanical chassis, motion system, embedded controller, power distribution system, safety and security systems. • 4 degrees-of-freedom robotic arm mounted on the AMR. • Sequence of movements of cobots. • End effector suitable for handling pieces with characteristics to be defined by the customer. • Remote control panel for operation and status display. • Vision system for pieces location. • Control system of position of a 3-meter-long 7th axis.		
Schedule	Start Date	August 10, 2026	
	End Date	December 4, 2026	
Time Reporting	The planning will be done based on key dates instead of the effort and man hours needed to complete a task. Hence, progress tracking of the project will be done through deliverables and milestones.		

Cost Estimate			
Costing	Material and components: \$15,000 MXN Engineering work: Not billable Tooling and workshop: Supplied by sponsor		
Milestones			
September 4 th , 2026	Detailed design for all the subsystems, reviewed and accepted by the stakeholders: bill of materials, electric schematics, CAD model, control algorithms and software architecture.		
September 25 th , 2027	Manufacture and assembly of the main mechanical components of the AMR and robotic arm.		
October 16 th , 2026	Delivery of electronics subsystems: embedded controller, power stages.		
November 13 th , 2026	Integration of control software with mechanical and electronic stages.		
November 27 th , 2026	Integration and validation of complete system of pieces location and handling.		
Assumptions, Constraints, Dependencies, Impacts, and Risks			
Assumptions	Material supplied by sponsor: motors, drivers, encoders, battery, PLA and TPU filament. Access to sponsor's tools, workshops, and laboratories. Final list of requirements by August 25th, 2026. Fully functional development environment. Reuse of mechanical chassis of Automated Guided Vehicle developed in 4th semester. Conditions of operation environment are fixed and controlled.		
Constraints	Use of off-the-shelf full solutions for AMR or robotic arm is not allowed. Fixed due date: December 4th, 2026.		
Risks	Supplier's delivery dates in conflict with project's milestones. Cost of project beyond estimated cost. Damage or major failures near due date. Limited access to sponsor's facilities and tools.		
Name/Signatures		Roles and Responsibility	
Ixchel Orozco Viveros		Project manager / Mechanical design	
Iván Delgado Ramos		Software developer / electronics design	
Sebastián Rivera López		Mechanical design	
Emilio Medina Velazquez		Software developer / electronics design	
Gerardo Galaviz Becerra		Mechanical design	
We agree that this is a viable project. We authorize the beginning of the project planning stage.			
Signature		Signature	
Date		Date	
Project Sponsor		Senior Manager	